

Task 3: Develop a C program using fork() that creates a parent and child process. Display the Process ID (PID), Parent Process ID (PPID), and process states at different stages of execution.

Flow

Chart

```
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <stdlib.h>

int main()
{
    pid_t pid;

    printf("Before fork():\n");
    printf("Parent Process PID: %d\n", getpid());
    printf("Parent Process PPID: %d\n\n", getppid());

    pid = fork();

    if (pid < 0)
    {
        perror("fork failed");
        return 1;
    }

    else if (pid == 0)
    {
        printf("CHILD PROCESS:\n");
        printf("Child PID: %d\n", getpid());
        printf("Child PPID: %d\n", getppid());
        printf("Child Process State: Running\n\n");

        sleep(5);

        printf("Child Process State: Terminated\n");
        exit(0);
    }
}
```

```

else
{
    printf("PARENT PROCESS\n");
    printf("PID: %d\n", getpid());
    printf("Child PID: %d\n", pid);
    printf("Parent is waiting for child...\n");

    wait(NULL);

    printf("Child process has terminated.\n");
    printf("Parent process is terminating.\n");
}

return 0;
}

```

```

ramya@LAPTOP-OPDAF309: ~$ mkdir week3
ramya@LAPTOP-OPDAF309: ~$ cd week3
ramya@LAPTOP-OPDAF309: ~/week3$ nano task3.c
ramya@LAPTOP-OPDAF309: ~/week3$ gcc task3.c -o task3
ramya@LAPTOP-OPDAF309: ~/week3$ ./task3
Before fork():
Process ID (PID): 563
Parent Process ID (PPID): 318

PARENT PROCESS
PID: 563
Child PID: 564
Parent is waiting for child...
CHILD PROCESS
PID: 564
PPID: 563
Child is running...
Child process is terminating.
Child process has terminated.
Parent process is terminating.
ramya@LAPTOP-OPDAF309: ~/week3$ |

```

Task 4: Design an experiment to observe process state transitions (Ready, Running, Waiting, Terminated) using Linux monitoring tools such as ps, top, and /proc. Document the observations.

Flow Chart:

```

Parent is TERMINATING...
ramya@LAPTOP-OPDAF309:~/Week3$ ./process &
[1] 559
Parent process started
Parent PID : 559
Parent PPID : 333

--- PARENT PROCESS ---
Parent PID : 559
Child PID : 561
Parent is WAITING for child...

--- CHILD PROCESS ---
Child PID : 561
Child PPID : 559
Child is RUNNING...
ramya@LAPTOP-OPDAF309:~/Week3$ Child is TERMINATING...
Child terminated.
Parent is TERMINATING...

```

```

ps -f

```

UID	PID	PPID	C	STIME	TTY	TIME	CMD
ramya	333	332	0	10:04	pts/0	00:00:00	-bash
ramya	570	333	0	10:09	pts/0	00:00:00	ps -f

```

[1]+  Done                  ./process
ramya@LAPTOP-OPDAF309:~/Week3$ ps -o pid,ppid,state,stat,cmd

```

PID	PPID	S	STAT	CMD
333	332	S	Ss	-bash
584	333	R	R+	ps -o pid,ppid,state,stat,cmd

```
ramya@LAPTOP-OPDAF309:~/Week3$ pgrep process
ramya@LAPTOP-OPDAF309:~/Week3$ cat /proc/333/status
Name:      bash
Umask:     0022
State:     S (sleeping)
Tgid:      333
Ngid:      0
Pid:       333
PPid:      332
TracerPid: 0
Uid:       1000    1000    1000    1000
Gid:       1000    1000    1000    1000
FDSize:    256
Groups:    4 24 27 30 46 100 1000
NStgid:    333
NSpid:     333
NSpgid:    333
NSsid:     333
Kthread:   0
VmPeak:    6784 kB
VmSize:    6784 kB
VmLck:     0 kB
VmPin:     0 kB
VmHWM:     6068 kB
VmRSS:     6068 kB
RssAnon:           2204 kB
RssFile:           3864 kB
RssShmem:          0 kB
VmData:    2144 kB
VmStk:     132 kB
VmExe:     1036 kB
VmLib:     1908 kB
VmPTE:      56 kB
VmSwap:     0 kB
HugetlbPages:      0 kB
CoreDumping: 0
THP_enabled: 1
untag_mask: 0xffffffffffffffff
Threads:    1
```

```

SigQ: 1/30983
SigPnd: 000000000000000000
ShdPnd: 000000000000000000
SigBlk: 00000000000010000
SigIgn: 00000000000384004
SigCgt: 000000004b813efb
CapInh: 000000000000000000
CapPrm: 000000000000000000
CapEff: 000000000000000000
CapBnd: 000001ffffffffffff
CapAmb: 000000000000000000
NoNewPrivs: 0
Seccomp: 0
Seccomp_filters: 0
Speculation_Store_Bypass: thread vulnerable
SpeculationIndirectBranch: conditional enabled
Cpus_allowed: fff
Cpus_allowed_list: 0-11
Mems_allowed: 00000000,00000000,00000000,00000000,00000000,00000000,000000
00,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,0
00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000
000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000001
Mems_allowed_list: 0
voluntary_ctxt_switches: 452
nonvoluntary_ctxt_switches: 3

```

```

ramya@LAPTOP-OPDAF309:~/Week3$ grep -E "Name|State|Pid|PPid" /proc/333/statu
s
Name: bash
State: S (sleeping)
Pid: 333
PPid: 332
TracerPid: 0

```

```

ramya@LAPTOP-OPDAF309:~/Week3$ top
top - 14:55:01 up 13 min,  1 user,  load average: 0.07, 0.04, 0.00
Tasks:  24 total,   1 running,  23 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,   0.0 sy,   0.0 ni,100.0 id,   0.0 wa,   0.0 hi,   0.0 si,   0.
MiB Mem :  7756.6 total,   7139.4 free,   553.9 used,   214.3 buff/cache
MiB Swap:  2048.0 total,   2048.0 free,    0.0 used.  7202.7 avail Mem

```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+
1	root	20	0	24176	15440	11636	S	0.0	0.2	0:01.15
2	root	20	0	3180	2204	2072	S	0.0	0.0	0:00.01
7	root	20	0	3180	2048	1940	S	0.0	0.0	0:00.00
50	root	19	-1	50404	16812	15508	S	0.0	0.2	0:00.24
82	systemd+	20	0	22416	14484	11992	S	0.0	0.2	0:00.09
89	root	20	0	35380	12320	9212	S	0.0	0.2	0:00.72
117	root	20	0	2888	1952	1820	S	0.0	0.0	0:00.05
123	root	20	0	4472	3152	2900	S	0.0	0.0	0:00.02
129	message+	20	0	8792	5368	4692	S	0.0	0.1	0:00.10
156	root	20	0	44096	29824	14684	S	0.0	0.4	0:00.29
160	root	20	0	18448	9424	8168	S	0.0	0.1	0:00.14
190	syslog	20	0	220548	5828	4536	S	0.0	0.1	0:00.08
225	_chrony	20	0	23908	11312	7092	S	0.0	0.1	0:00.11
233	_chrony	20	0	12096	2404	1760	S	0.0	0.0	0:00.01
244	root	20	0	5492	2768	2584	S	0.0	0.0	0:00.01
268	root	20	0	123460	32564	17948	S	0.0	0.4	0:00.22
331	root	20	0	3184	1108	980	S	0.0	0.0	0:00.00
332	root	20	0	3200	1248	1108	S	0.0	0.0	0:00.11
333	ramya	20	0	6784	6068	3864	S	0.0	0.1	0:00.15
334	root	20	0	8648	5608	4764	S	0.0	0.1	0:00.02
385	ramya	20	0	22756	13408	10960	S	0.0	0.2	0:00.17
387	ramya	20	0	22916	4124	2124	S	0.0	0.1	0:00.00
413	ramya	20	0	6136	5476	3928	S	0.0	0.1	0:00.03
685	ramya	20	0	8172	6008	3888	R	0.0	0.1	0:00.14

```

ramya@LAPTOP-OPDAF309:~/Week3$ ps -eo pid,ppid,state,stat,cmd
  PID    PPID S  STAT CMD
    1         0 S  Ss  /sbin/init
    2         1 S  Sl+  /init
    7         2 S  Sl+  plan9 --control-socket 7 --log-level 4 --server-fd 8
   50        1 S  S<s  /usr/lib/systemd/systemd-journald
   82        1 S  Ss   /usr/lib/systemd/systemd-resolved
   89        1 S  Ss   /usr/lib/systemd/systemd-udev
  117        1 S  Ss   /bin/sh /usr/lib/systemd/scripts/chronyd-starter.sh -
  123        1 S  Ss   /usr/sbin/cron -f -P
  129        1 S  Ss   @dbus-daemon --system --address=systemd: --nofork --n
  156        1 S  Ss   /usr/bin/python3 /usr/bin/networkd-dispatcher --run-s
  160        1 S  Ss   /usr/lib/systemd/systemd-logind
  190        1 S  Ssl  /usr/sbin/rsyslogd -n -iNONE
  225       117 S  S    /usr/sbin/chronyd -n -F 1 -x
  233       225 S  S    /usr/sbin/chronyd -n -F 1 -x
  244         1 S  Ss+  /usr/sbin/agetty --noreset --noclear --issue-file=/et
  268         1 S  Ssl  /usr/bin/python3 /usr/share/unattended-upgrades/unatt
  331         2 S  Ss   /init
  332       331 S  S    /init
  333       332 S  Ss   -bash
  334         2 S  Ss   login -- ramya
  385        1 S  Ss   /usr/lib/systemd/systemd --user
  387       385 S  S    (sd-pam)
  413       334 S  Ss+  -bash
  765       333 R  R+   ps -eo pid,ppid,state,stat,cmd
ramya@LAPTOP-OPDAF309:~/Week3$ ps -o pid,ppid,state,cmd
  PID    PPID S  CMD
  333       332 S  -bash
  775       333 R  ps -o pid,ppid,state,cmd
ramya@LAPTOP-OPDAF309:~/Week3$ pgrep process
ramya@LAPTOP-OPDAF309:~/Week3$ ps -o pid,ppid,state,cmd | grep process
  796       333 S  grep --color=auto process

```